

EXPERIMENT 1 – BASIC SEARCH STRATEGIES

1(a) 8-Puzzle – Output

[[1, 2, 3, 4, 0, 5, 6, 7, 8],
[1, 2, 3, 4, 5, 0, 6, 7, 8],
[1, 2, 3, 4, 5, 8, 6, 7, 0],
[1, 2, 3, 4, 5, 8, 6, 0, 7],
[1, 2, 3, 4, 5, 8, 0, 6, 7],
[1, 2, 3, 0, 5, 8, 4, 6, 7],
[1, 2, 3, 5, 0, 8, 4, 6, 7],
[1, 2, 3, 5, 6, 8, 4, 0, 7],
[1, 2, 3, 5, 6, 8, 4, 7, 0],
[1, 2, 3, 5, 6, 0, 4, 7, 8],
[1, 2, 3, 5, 0, 6, 4, 7, 8],
[1, 2, 3, 0, 5, 6, 4, 7, 8],
[1, 2, 3, 4, 5, 6, 0, 7, 8],
[1, 2, 3, 4, 5, 6, 7, 0, 8],
[1, 2, 3, 4, 5, 6, 7, 8, 0]

1(b) 8-Queens – Output

[0, 4, 7, 5, 2, 6, 1, 3]
[0, 5, 7, 2, 6, 3, 1, 4]
[0, 6, 3, 5, 7, 1, 4, 2]
[0, 6, 4, 7, 1, 3, 5, 2]
...
[6, 4, 2, 0, 5, 7, 1, 3]
[7, 1, 3, 0, 6, 4, 2, 5]
[7, 1, 4, 2, 0, 6, 3, 5]
[7, 2, 0, 5, 1, 4, 6, 3]
[7, 3, 0, 2, 5, 1, 6, 4]

1(c) Cryptarithmic – Output

Cryptarithmic Solution Mapping:

{'Y': 2, 'M': 1, 'S': 9, 'N': 6, 'O': 0, 'E': 5, 'D': 7, 'R': 8}

EXPERIMENT 2 – A* AND MEMORY-BOUNDED A*

2(a) A* Search – Output

Enter the goal state: K

Enter the maximum depth limit: 4

2(b) A* and SMA* – Output

A* Path: ['A', 'B', 'E', 'G']

A* Cost: 6

SMA* Path: ['A', 'B', 'E', 'G']

SMA* Cost: 6

EXPERIMENT 3 – MINI-MAX ALGORITHM WITH ALPHA-BETA PRUNING

The optimal value is : 5

EXPERIMENT 4 – CONSTRAINT SATISFACTION PROBLEM

Solution:

csc -> Monday

maths -> Tuesday

phy -> Wednesday

che -> Monday

tam -> Monday

eng -> Tuesday

bio -> Wednesday

EXPERIMENT 5 – PROPOSITIONAL MODEL CHECKING

TT-Entails Evaluation (Does KB entail Q?): True

EXPERIMENT 6 – FORWARD CHAINING, BACKWARD CHAINING AND RESOLUTION

Forward Chaining:

Goal Found: True

Backward Chaining:

Goal Found: True

Resolution:

Resolved Clause: {'Q'}

EXPERIMENT 7 – NAÏVE BAYES CLASSIFIER

Naive Bayes Classification: [np.int64(0), np.int64(1)]

EXPERIMENT 8 – BAYESIAN NETWORKS AND INFERENCE

Probability of Rain = 0.2

Probability of Wet Grass = 0.26

Probability of Rain given Wet Grass = 0.69

EXPERIMENT 9 – GOAL-BASED AUTONOMOUS AGENT

Start State : 2

Goal State : 7

Path:

2 -> 3 -> 4 -> 5 -> 6 -> 7

Goal Reached Successfully!

EXPERIMENT 10 – REFLEX AGENT AND MODEL-BASED AGENT

----- **Reflex Agent** -----

Enter room status (Clean/Dirty): Dirty

Action: Clean the room

----- **Model-Based Agent** -----

Enter room name (A/B): A

Enter room status (Clean/Dirty): Clean

Memory: {'A': 'Clean'}

Action: Room is already clean

EXPERIMENT 11 – ENVIRONMENT-AWARE INTELLIGENT AGENT

Environment : Rain

Agent Action: Take an Umbrella

Another Example:

Environment : Cold

Agent Action: Wear a Jacket

EXPERIMENT 12 – LEARNING AGENT WITH MEMORY UPDATE

1. Learn

2. Recall

3. Exit

Enter your choice: 1

Enter Situation: Rain

Enter Action: Take Umbrella

Memory Updated!

1. Learn

2. Recall

3. Exit

Enter your choice: 2

Enter Situation: Rain

Known Action: Take Umbrella

1. Learn

2. Recall

3. Exit

Enter your choice: 3

Exiting...

EXPERIMENT 13 – COOPERATIVE MULTI-AGENT COMMUNICATION SYSTEM

Agent A sends to Agent B: Complete the assigned task

Agent B received from Agent A: Complete the assigned task

Agent B: Task acknowledged.

EXPERIMENT 14 – TRUST AND REPUTATION SYSTEM

Ratings: [5.0, 4.0, 5.0, 4.0, 5.0]

Average Rating: 4.6

Reputation: Excellent

EXPERIMENT 15 – DISTRIBUTED PROBLEM-SOLVING AGENTS

Agent A processed [10, 20, 30] -> Sum = 60

Agent B processed [40, 50, 60] -> Sum = 150

Final Sum = 210